Global Context-Augmented Objection Detection in VHR Optical Remote Sensing Images

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***Abstract*— The deep learning method, especially convolution neural networks (CNNs), has recently made ground-breaking advances on object detection in very-high-resolution (VHR) optical remote sensing images. However, as CNN is originally designed for the classification of natural images, these methods are not very suitable for object detection of remote sensing images. First, current CNN-based approaches have difficulty to deal with objects that have large rotation variation, which is widely existed in optical remote sensing images. Second, the detectors based on CNN only have limited receptive fields and, thus, can hardly utilize the global contextual information that is essential for accurate detection of small targets. To address these two issues, this article proposes a novel deep learning-based object detection framework, including a geometric transform module (GTM) and a global contextual feature fusion module (GCFM). Especially, the GTM combines rotation and flip transformation to deal with the multiangle characteristics of objects. The GCFM uses a spatial attention mechanism to adaptively involve global contextual information in feature maps to improve the recognition and location accuracy of targets. We introduce the two modules into the YOLOv3 framework to achieve end-to-end detection with high performance and efficiency. Comprehensive evaluations on three publicly available object detection data sets demonstrate the excellent performance of the proposed methods.**

***Index Terms*— Attention mechanism, global contextual information, object detection, remote sensing images, rotation invariant.**

I. Introduction

A LARGE number of remote sensing images with the high spatial resolution are available as the development of remote sensing image sensors and space technology. The imagery facilitates a wide range of applications such as segmentation [1], [2], unmixing [3]–[5], data fusion [6]–[8], pansharpening [9]–[11], object detection [12]–[14], and classification [15]–[17]. Recently, object detection in optical remote sensing images has gained increasing interest as its importance in land planning, urban monitoring, and traffic planning [18]–[20]. Traditional object detection methods often start from generating region of interests (ROIs) by grouping pixels into clusters and detect objects based on handcraft image features, such as SIFT and HOG [21]–[27]. In recent years, due to the rapid advance of the machine learning technique, particularly deep learning methods, many approaches begin to use multilayer convolution neural networks (CNNs) [28]–[31]. The region-based convolution network (R-CNN) achieves excellent object detection accuracy by using convolution networks to generate the feature of remote sensing images [32]. Compared with traditional R-CNN and its accelerated version SPPnet [33], Fast R-CNN [34] enables end-to-end detector training by using a multitask loss on a CNNs and shows compelling accuracy and speed. Based on Fast R-CNN, Faster R-CNN [35] introduces region proposal networks (RPNs) to generate the ROIs, which enables a unified, deep-learning-based object detection system to run at near-real-time frame rates. All the abovementioned models need to generate ROIs, and then using the ROIs to predict the bound box and the classification results of the object, this kind of model could be categorized as the two-stage or region-based methods. Other kinds of models are one-stage or region-free methods; they treat object detection as a single regression problem and directly generate bounding boxes and classification results simultaneously. Compared with two-stage methods, one-stage methods could get almost real-time detection but with lower accuracy. Methods of this kind include YOLO [36]–[38] and SSD [39], [40].

Based on the great success of object detection algorithms, researchers present innovative structures and better solutions based on CNNs for object detection in remote sensing images. Han *et al.* [41] combined weakly supervised learning that requires only binary labels and high-level feature learning by using deep Boltzmann machines in object detection of remote sensing images. R-P-Faster R-CNN [42] adds a pretraining mechanism to improve the efficiency of the multiclass geospatial object detection by transfer learning from the natural imagery domain to the remote sensing imagery domain. Long *et al.* [43] constructed a new localization framework that uses a combination model of CNN and an unsupervised

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score-based bounding box regression (USB-BBR) algorithm to improve the object localization accuracy in high-resolution remote sensing images. Zou and Shi [44] proposed random access memories (RAMs) and integrated small target prior to the CNN-based object detection paradigm. Guo *et al.* [45] used a unified multiscale CNN for detection of multiscale objects in high-resolution satellite images. Chen *et al.* [46] introduced semantic segmentation-aware CNN features and global activation blocks to the single-shot detection framework to improve positioning accuracy on relatively small objects. Zhang *et al.* [47] proposed a scale adaptive proposal network (SANet) to improve the accuracy of multiobject detection in remote sensing images. Dong *et al.* [48] leveraged a technique of Sig-NMS and use transfer learning to increase the detecting performance of small targets. Zhou *et al.* [13] proposed Fisher regularization that could make full use of hidden knowledge for efficient knowledge transferring.

However, although these CNN-based models have achieved great success, various challenges still exist in the remote sensing image object detection field, such as the complex and varied appearances of the geospatial object, the complex background, and the difficulty in fast detecting the large scene image. Two practical problems stand in their way. First, remote sensing images are taken from the upper perspective, and the objects usually have diverse rotation variations. However, it is difficult for the CNN-based method to effectively handle large rotation variations as the CNN architecture is specifically designed for the classification of nature scene images in which objects have small orientation variations. Second, object detection in remote sensing images becomes more complicated because of the diversity and complexity of remote sensing images. For example, the object and the background may have a similar texture and color features. Furthermore, different objects of geospatial objects could be similar in appearance. To address these two issues, rotation-invariant and contextual information could play a fundamental role.

Many researchers have shown that introducing the rotation- and contextual information could improve detection performance. RICNN [49] introduces a new rotation-invariant layer on the basis of the AlexNet CNN to achieve rotation-invariance. Li *et al.* [50] introduced additional multiangle anchors to deal with the multiangle characteristics. Ren *et al.* [51] proposed a deformable Faster R-CNN that is constructed by substituting the standard convolution layer with a deformable convolution layer in the last network stage to model geometric transformations. Zhou *et al.* [12] utilized an encoder–decoder architecture, rotated feature network (RFN) to produce rotation-sensitive feature maps for regression and rotation-invariant feature maps for classification. ORSIm [52] introduces a spatial-frequency channel feature, which is composed of color and gradient magnitude channels from the spatial domain and rotation invariant features from the frequency domain. FRIFB [53] uses a mathematically rotation-invariant descriptor (FourierHOG) against rotation behaviors of arbitrary continuous angle for the task of geospatial object detection.

Kang *et al.* [54] introduced the contextual information by using features around the ROI. Li *et al.* [50] used a local-contextual feature fusion network to combine the local and contextual features to form a powerful joint representations. Ma *et al.* [14] leveraged a contextual information fusion subnetwork to fuse both local contextual features and object–object relationship contexual features. However, most of the CNN-based detection methods only use local contextual information, e.g., the contextual feature used here only included the information surround the ROI region. The contextual information far away from the ROI region is still not used. It is difficult for CNN to effectively make use of the background or contextual information because the effective receptive field is local.

To effectively address the aforementioned issues, we introduced two modules: geometric transform module (GTM) and global-contextual feature fusion module (GCFM), which could be used in the CNN-based geospatial object detection models. The GTM uses a simple and effective data augmentation technique to train a rotation-insensitive CNN model for geospatial object detection. The GCFM consists of a position embedding layer and a spatial attention layer, both of which could be effectively integrated into CNN feature maps. The spatial attention layer explores attention mechanism [55]–[58] to involve the contextual information to each of the detection point. To the best of our knowledge, it is the first time that the attention mechanism is used in a geospatial object detection framework to adaptively learn and fuse the spatial information in the feature map. By combining the GTM and GCFM with the object detection model YOLOv3, we demonstrated the superiority and effectiveness of our proposed method (YOLOv3-Att) on three data sets. Our contributions can be summarized into four components.

1) We proposed a novel, efficient deep learning-based geospatial object detection framework (YOLOv3-Att) by integrating GTM and GFCM with YOLOv3. This framework is robust to rotation variant and strong to detect an object with diversity appearance.
2) A GTM is designed by randomly applying flip or rotation transformation to training images to handle various object deformation behaviors in remote sensing imagery.
3) We proposed GCFM that uses spatial attention mechanism to utilize the global contextual information to build powerful joint representations for object detection.
4) We comprehensively analyze the attention weight learned by the model and found how the global-contextual information help to improve the detection performance.

The rest of this article is organized as follows. Section II presents the details of the proposed methods. Section III compares the results of our method with other models on three optical remote sensing image data sets. Section IV analyzes the effects of the proposed model on detection performance. Finally, we end this article with conclusions in Section V.

II. Overview of Proposed Method

Fig. 1 illustrates the overall framework of our proposed method. Our proposed method is based on the single-stage object detection model of YOLOv3. Especially, given a remote

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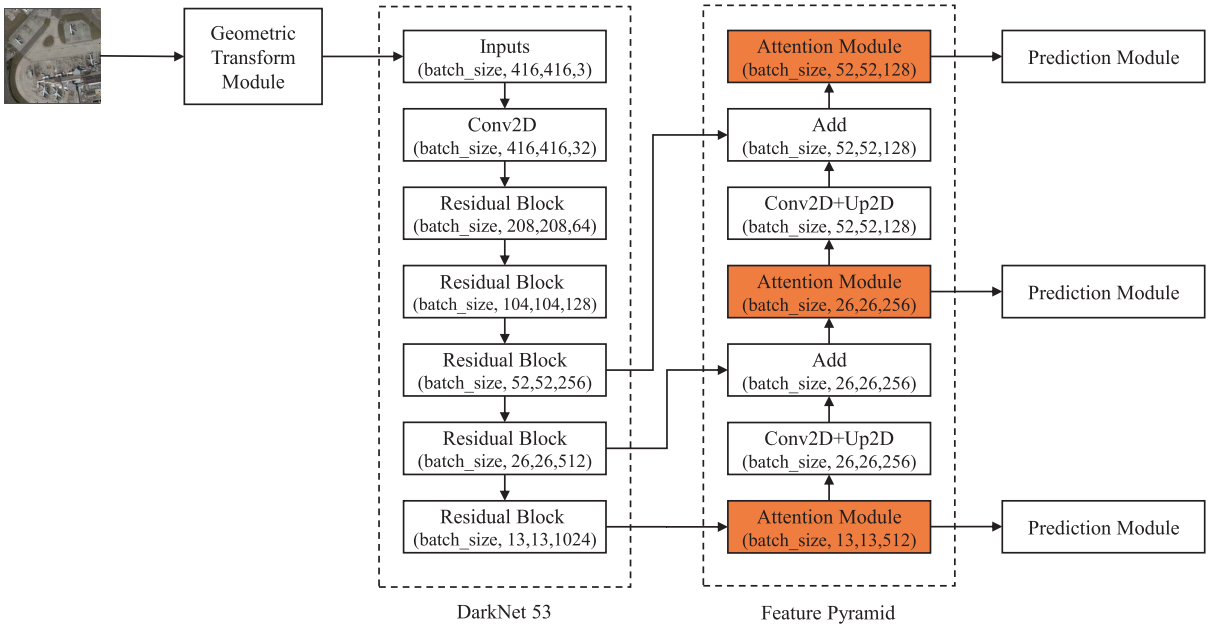


Fig. 1. Specific framework of our proposed method. The framework consists of four parts: GTM, DarkNet53, feature pyramid, and prediction module. The feature pyramid is constructed with GCFM. The "Up2D" represents the upsampling operation, and "Add" is the elementwise add operation.

sensing image as input, we first use GTM to generate multiangle images to handle the problem of rotation variant of geospatial objects. Second, the DarkNet [38] is used as the backbone to extract features. Third, to enable the model to detect multiscale objects in the images, we utilize a feature pyramid to generate multilevel feature maps. Then, a GCFM based on attention mechanism is proposed to build a joint feature map by using each multiscale feature map. Finally, a prediction module uses the joint representation to output the confidence score and generate bounding box regression parameters of the objects. We next describe each individual module of the proposed framework.

A. GTM

Traditional CNN-based object detection models only consider multiple scale and multiple aspect ratio characteristics of objects and, thus, lack the ability to handle the large rotation characteristics of objects. However, as remote sensing images are taken from the top view by the spaceborne or airborne sensor, objects in these images often contain various rotation angles. In order to make the CNN-based object detection model more robust to rotation transformation, we added a GTM after the input layer of the model. Especially, the GTM mainly includes two operations: random rotation transformation and random flip transformation. The random rotation transformation rotates the input image by a certain angle. Due to the existence of multiple rotation angles of the target in the image, the bounding boxes of the objects are rectangles that are represented by the coordinates of their top left and bottom right. In order to get accurate coordinates of bounding boxes of the objects in the rotated image, we only choose three rotation angles, i.e., 90°, 180°, and 270°. Random flip transformation flips the image horizontally (top–bottom flip) or vertically (left–right flip) randomly. The geometric transformation module randomly selects one transformation at a time to rotate or flip the image accordingly.

TABLE I
COORDINATES OF THE BOUNDING BOX AFTER THE TRANSFORMATION

Transformation	Upper left corner	Lower right corner
Original	(x_1, y_1)	(x_2, y_2)
90° rot	$(y_1, w - x_2)$	$(y_2, h - x_1)$
180° rot	$(w - x_2, h - y_2)$	$(w - x_1, h - y_1)$
270° rot	$(h - y_2, x_1)$	$(h - y_1, x_2)$
Top-bottom flip	$(x_1, h - y_2)$	$(x_2, h - y_1)$
Left-right flip	$(w - x_2, y_1)$	$(w - x_1, y_2)$

After the transformation of the image, we should recalculate the coordinates of the bounding boxes because the origin coordinates are always at the top left corner of the image. Suppose that the top left and bottom right coordinates of the bounding boxes in the original image are (x_1, y_1) and (x_2, y_2). Table I shows the calculation formula of the coordinates after transformations.

B. DarkNet and Feature Pyramid

The DarkNet53 network [38] is utilized as the backbone for extracting features. DarkNet53 has mainly five residual blocks, each of which consists of successive 3×3 and 1×1 convolution layers and shortcut connections. The scale of the feature map in each residual block is decreasing, and the number of channels is increasing. Then, the final feature maps from the last three residual blocks are used to build a feature pyramid. Especially, the feature maps with lower resolution are upsampled and added to the feature map with higher

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resolution. To augment the feature with global contextual information, a GCFM is introduced in the building process of the feature pyramid, as shown in Fig. 1.

C. GCFM

Remote sensing images are usually large scale and with high resolution, which means that every location in the images may contain useful information. Furthermore, many different kinds of geospatial objects are similar to the appearance in remote sensing images. In remote sensing image object detection, the context could play an important role in the model to better identify and locate targets. However, as the effective receptive field [59] of CNNs only cover a small area, most of the existing CNN-based object detection algorithms only use local contextual information. We propose a GCFM to fuse global contextual information in the feature map by using an attention mechanism in the spatial dimension.

The attention mechanism is first introduced in RNN encoder–decoder for the statistical machine translation problem [60]. RNN encoder–decoder consists of two RNNs that act as an encoder and a decoder pair. The encoder maps a variable-length source sequence to a fixed-length vector, which is named context vector. The decoder maps the context vector back to a variable-length target sequence. It is straightforward to use the last hidden state of the encoder as the context vector [60]. However, only use the last hidden state as the context vector makes it difficult for the model to cope with long sentences. The attention mechanism is proposed to allow the model to automatically search for parts of a source sentence that are relevant to the target word, without having to form these parts like a hard segment explicitly. The attention mechanism uses the weighted average of multiple hidden states to generate the context vector. The weight of each hidden state is named attention weight or alignment. Bahdanau *et al.* [58] used the hidden states of encoder and decoder as the input of a single-layer multilayer perceptron to generate the alignment. Chorowski *et al.* [56] improved the attention mechanism and introduce a hybrid attention mechanism that is based both on location and content. Britz *et al.* [57] proposed to use the inner product of hidden states of encoder and decoder instead of using multilayer perceptron to generate the alignment. Recently, researchers propose self-attention or intra-attention, an attention mechanism that uses different positions of a single sequence as the input to generate a representation of the sequence. Transformer [55] is a successful transduction model that relying entirely on self-attention to compute representations of input and output without using sequence aligned RNNs or convolution. This work uses self-attention to generate global information to build powerful joint representation for object detection in remote sensing images. Compared with the traditional attention mechanism that is usually applied in the temporal dimension, we use the attention mechanism to explore spatial relationships between each location.

Fig. 2 shows building block of the GCFM. The GCFM is mainly constituted of three parts. The first part is the position embedding layer that injects information about the absolute or relative position of each pixel to the feature

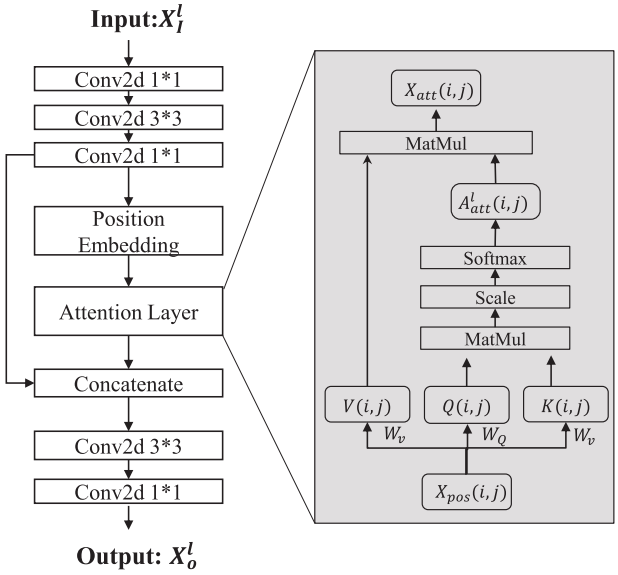


Fig. 2. Structure of attention block used to fuse global-contextual information. Here, MatMul represents the matrix multiplication operation, and scale represents the scale operation.

map. There are basically two ways to do position embedding: fixed positional embedding by positional encoding [55] and learned positional embedding [61]. In this article, we use the learned positional embedding follows [61].

Suppose that $X_I^l \in \mathbb{R}^{h_l \times w_l \times c_l}$ is the lth feature map of the feature pyramid. First, three convolution layers are appended on X_I^l to generate the feature map X^l. Then, we inject some information about the relative or absolute position to X^l using position embedding. The feature map after position embedding X_{pos}^l is

$$X_{\text{pos}}^l = X^l + W_{\text{pos}}^l \tag{1}$$

in which W_{pos}^l is the position embedding matrix. W_{pos}^l is the parameter matrix that has the same shape as X^l and is randomly initialized before the training.

The second part of GCFM is the attention fusing layer. Let us use $X_{\text{pos}}^l(i, j)$ to denote the c_l-dimension feature vector in feature map X_{pos}^l. Index (i, j) is the spatial position in the feature map, $1 \le i \le h_l, 1 \le j \le w_l$. Attention fusing layer using self-attention mechanism to fuse feature $X_{\text{pos}}^l(i, j)$ with features at other locations to get $X_{\text{att}}^l(i, j)$. Especially, we first use linear projection to convert the feature vector $X_{\text{pos}}^l(i, j)$ to three different vectors that we denote as $Q_{\text{pos}}^l(i, j)$, $K_{\text{pos}}^l(i, j)$ and $V_{\text{pos}}^l(i, j)$, defined by

$$\begin{aligned} Q_{\text{pos}}^l(i, j) &= X_{\text{pos}}^l(i, j) \times W_Q \\ K_{\text{pos}}^l(i, j) &= X_{\text{pos}}^l(i, j) \times W_K \\ V_{\text{pos}}^l(i, j) &= X_{\text{pos}}^l(i, j) \times W_V \end{aligned} \tag{2}$$

where $W_Q \in \mathbb{R}^{c_l \times d_q}, W_K \in \mathbb{R}^{c_l \times d_k}$, $W_V \in \mathbb{R}^{c_l \times d_v}$ are the trainable projection matrix, and $\times$ represents the matrix multiplication operation. We follow the settings in [55] and employ $d_q = d_v = d_k = c_l/r$. Here, r is the ratio between the dimension of the projection vector and the feature map

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channels. The attention weight or the alignment is calculated by $Q^l_{\text{pos}}(i,j)$ and $K^l_{\text{pos}}(i,j)$

$$T^l_{\text{att}}(i,j)(k,l) = \frac{Q^l_{\text{pos}}(i,j)\cdot K^l_{\text{pos}}(k,l)}{\sqrt{d_k}} \tag{3}$$

$$A^l_{\text{att}}(i,j)(k,l) = \text{softmax}(T^l_{\text{att}}(i,j)(k,l)) = \frac{\exp(T^l_{\text{att}}(i,j)(k,l))}{\sum_{k,l}(T_{\text{att}}(i,j)(k,l))} \tag{4}$$

in which $\cdot$ is the dot-product. $A^l_{\text{att}}(i,j)(k,l)$ represents the attention weight between feature vector $X^l_{\text{pos}}(i,j)$ and $X^l_{\text{pos}}(k,l)$. d_k is the dimension of vector $K^l_{\text{pos}}(k,l)$. The output of the attention fusing layer is the weighted sum of V^l_{pos}

$$X^l_{\text{att}}(i,j) = \sum_{k,l} A^l_{\text{att}}(i,j)(k,l)V^l_{\text{pos}}(k,l) \tag{5}$$

where $X^l_{\text{att}}(i,j)$ is the fused feature vector with global contextual information, which means that $X^l_{\text{att}}(i,j)$ contains information of every position of the feature map. It is worth noticing that the attention weight is generated by using multiplicative (dot-product) attention, which is much faster and more space-efficient in practice than additive attention. The scale factor $(d_k)^{1/2}$ is used to avoid the product from growing too large in magnitude and push softmax function into the region where it has extremely small gradients.

The final part is the lateral connection. After the attention layer, we add a lateral connection to the feature fusion module. The lateral connection concatenates the original feature X^l with the attention fused feature $X^l_{\text{att}}(i,j)$. Then, two convolution layers are used to generate the final feature map X^l_o, which is aimed to reduce the aliasing effect and reduce channel dimensions. The final combined feature map X^l_o contains both position information and the global contextual information. All these layers are designed to guarantee the size of feature maps unchanged.

D. Prediction Module

After GCFM, the feature map is used to predict the bounding boxes of objects in remote sensing images. The prediction module predicts bounding boxes for each location at the feature map by using anchor boxes. Anchor boxes are predefined boxes that have different shapes and scales. Here, we use bounding box clustering [37] to generate 12 anchor boxes. The anchor boxes are then divided into three categories that are assigned to corresponding pyramid feature maps to generate bounding box predictions. For each anchor box at every position in the feature map, the network predict four bounding-box coordinates offsets t_x, t_y, t_w, and t_h, one objectness score t_o, and a bounding-box class probabilities t_c. If the anchor box at position c_x, c_y has the width and height p_w, p_h, the corresponding prediction could be calculated by offsets using the formula

$$\begin{aligned} b_x &= s(t_x + c_x) \\ b_y &= s(t_y + c_y) \\ b_w &= p_w e^{t_w} \\ b_h &= p_h e^{t_h} \end{aligned} \tag{6}$$

where scale is the strides of the feature map with respect to the input image, and b_x and b_y are the central coordinates of the prediction bounding-box. b_w and b_h are the width and height of the prediction bounding-box. In the model, t_x and t_y are the location coordinates of the bounding box relative to the location of the grid cell. The prediction confidence score t_s of each bounding-box is calculated by

$$t_s = t_o * t_c \tag{7}$$

where $*$ is the scalar multiplication. The prediction confidence score could be used to select the bounding-box and perform nonmaximum suppression (NMS) [37].

As for training, the model uses a multipart loss function, which can be represented as

$$L = L_{\text{conf}} + L_{\text{reg}} + L_{\text{cls}}. \tag{8}$$

As shown in 8, the loss function is consisted with three terms. L_{conf} is the loss for the objectness score, which is formulated as

$$L_{\text{conf}} = \sum_{i\in\mathbb{I}_{\text{pos}}} L_b(t_c(i),1) + \sum_{i\in\mathbb{I}_{\text{neg}}} L_b(t_c(i),0) \tag{9}$$

where i is the index of anchor box in a minibatch, and $t_c(i)$ is the predicted objectness score of the predicted box associated with ith anchor box. $\mathbb{I}_{\text{pos}}$ is the index set of positive anchor box, and $\mathbb{I}_{\text{neg}}$ is the index set of negative anchor box. In the training phase, an anchor box is positive only if it overlaps a ground-truth object by more than any other anchor boxes. The negative anchor box is not the best match and has an intersection over union (IOU) overlap lower than 0.5 with any ground-truth box. L_b is the log loss function over two classes. L_{cls} is the classification loss, which is defined as

$$L_{\text{cls}} = \sum_{i\in\mathbb{I}_{\text{pos}}} L_b(t_c(i), t^*_c(i)) \tag{10}$$

where $t_c(i)$ and $t^*_c(i)$ are the predict and ground-truth classes, respectively. L_{reg} is the bounding box regression loss, which is defined as

$$L_{\text{reg}} = \sum_{i\in\mathbb{I}_{\text{pos}}} \left[L_b\left(t_x(i), t^*_x(i)\right) + L_b\left(t_y(i), t^*_y(i)\right)\right. \tag{11}$$

$$\left. + L_s\left(t_w(i), t^*_w(i)\right) + L_s\left(t^*_h(i), t^*_h(i)\right)\right] \tag{12}$$

where L_s is the square loss function. It is worth noticing that only positive anchor boxes contribute to L_{cls} and L_{reg}.

III. Data Sets and Experiments

A. Data Description

To evaluate the performance of our proposed method, we conduct experiments on three remote sensing image data sets, i.e., NWPU Very High Resolution (VHR)-10 [62], RSOD [43], and UCAS-AOD [63]. Among them, NWPU VHR-10 is a data set labeled by Northwestern Polytechnical University and contains multisource and multiresolution target detection images. There are a total of 800 remote sensing images in the data set, of which 715 are from Google Maps with a spatial resolution of 0.5–2.0 m and the remaining 85 color-composited infrared images from Vaihingen with a resolution of 0.08 m.

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Among these 800 images, there are 650 images in which each image at least contains one object to be detected. The rest 150 images are the backgrounds that contain no object to be detected. There are ten different types of objects contained within the NWPU VHR-10 data set, i.e., airplanes, ships, storage tanks, baseball diamond, tennis courts, basketball courts, ground track field, harbor, bridges, and vehicles.

The RSOD data set is a remote sensing image target detection data set labeled by Wuhan University. The data set contains a total of 976 remote sensing images, with data from Google Maps and Tianditu. The objects in these images are labeled with four categories: airplane, playground, overpass, and oil tanks.

The UCAS-AOD data set is a high-resolution aerial object detection data set labeled by the University of Chinese Academy of Sciences. It is obtained by cropping with Google Earth software in some parts of the world. The data set consists of 1510 images (about 1000×1000 pixels) and two categories of cars and planes.

B. Evaluation Metrics

In the field of object detection, IOU is usually used as an assessment indicator of whether the detection result is correct. The IOU evaluates the degree of coincidence between the true bounding box B_{gt} and the predicted bounding box B_p based on the Jaccard index. The IOU is defined as

$$\text{IOU} = \frac{\text{area}(B_p \cap B_{gt})}{\text{area}(B_p \cup B_{gt})}. \tag{13}$$

Here, $\text{area}(B_p \cap B_{gt})$ represents the area of the intersection of B_p and B_{gt}; $\text{area}(B_p \cup B_{gt})$ is the area of union of B_p and B_{gt}. In the application, an overlap threshold σ_{iou} is given. If $\text{IOU} \geq \sigma_{\text{iou}}$, the predicted bounding box is considered to be true positive. Otherwise, the predicted bounding box is considered to be a false positive. The detection precision is defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

where the TP represents the number of true positive predicted bounding box and FP represents the number of false positive predicted bounding box. To evaluate the performance of a model, the percentage of total true objects correctly detected by the model should also be considered. The recall is defined as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{15}$$

in which FN is the number of objects that are not detected.

In order to comprehensively evaluate the performance of a model, it is reasonable to use the average of accuracy when taking different values of confidence thresholds σ_{conf}. Average precision (AP) is commonly used in object detection evaluation. It is defined as the average accuracy of the model at different recall levels. Here, we use the definition of AP that used in PASAL VOC 2012 challenge, which is defined as

$$\text{AP} = \sum_n (r_{n-1} - r_n)\rho_{\text{interp}}(r_{n+1}) \tag{16}$$

with

$$\rho_{\text{interp}}(r_{n+1}) = \max_{\tilde{r}:\tilde{r} \geq r_{n+1}} \rho(\tilde{r}) \tag{17}$$

where $\rho(\tilde{r})$ is the measured precision at recall $\tilde{r}$.

C. Experiment Settings

The backbone of the network was initialized using the weights pretrained on the COCO data set. In the training phase, Adam [64] was chosen as the optimizer. Limited by the amount of GPU memory, the number of images in a minibatch was set to 8. The training procedure was comprised of two stages. In the first stage, we have frozen the parameters of the backbone and trained the rest layers for 50 epochs using a learning rate of 0.001. In the second stage, the whole network was unfreeze and trained for another 350 epochs. The learning rate of the second stage was set to 0.0001 and decayed by a factor of 10 at epochs 120 and 240. The anchor boxes were determined by using k-means dimension clustering following [37]. We have chosen 12 clusters and sorted the anchor boxes by three scales that were assigned the corresponding feature map. For each data set, the split ratios of the training, validation, and test data were 60%, 20%, and 20%, respectively. The experiments were implemented on a work station with two Intel Xeon CPU E5-2670 at 2.60 GHz and a single GPU NVIDIA GTX 1080Ti. Deep learning framework Keras was used in our experiment for implementing our method.

D. Comparison

For evaluation, we used the YOLOv3 as the baseline and demonstrate the performance of our proposed method. The detection results are shown in Tables II–IV. Here, YOLOv3-Rot is YOLOv3 with the GTM only; YOLOv3-Att is YOLOv3 with the GTM and GCFM.

On the NWPU VHR-10 data set, we compared the proposed method with several state-of-the-art methods as follows: RICNN [62], R-FCN [65], Faster R-CNN [35], Deformable R-FCN [66], Deformable Faster R-CNN [51], Multi-Scale CNN [45], RDAS512 [46], R^2CNN++ [67], Improved Faster R-CNN [68], and Weighted Ensemble [69].

Our method YOLOv3-Att gets mAP of 94.49% which is better than all compared methods. Especially, our method obtains 2.74% and 2.18% performance gains, in terms of mean AP over all ten object categories, compared with the R^2CNN++ and Improved Faster R-CNN, respectively. Moreover, our method performs even better than the ensemble method [69].

Especially, for vehicle and bridge, we obtain significant performance improvement. It is worth noticing that almost all the compared methods use the two-stage detection framework; our method helps the one-stage method to achieve better performance. This demonstrates a high superiority of the proposed method compared with the existing state-of-the-art methods.

We further evaluate how the evaluation metric for GTM and GCFM influences the performance. As shown in Table II,

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TABLE II
EVALUATION RESULTS ON NWPU VHR-10 DATA SETS

	Airplane	Ship	Storage tank	Baseball diamond	Tennis court	Basketball court	Ground track field	Harbor	Bridge	Vehicle	mAP
RICNN	88.40	77.30	85.30	88.10	40.80	58.50	86.70	68.60	61.50	71.10	72.60
R-FCN	81.70	80.60	66.20	90.30	80.20	69.70	89.80	78.60	47.80	78.30	76.30
Faster R-CNN	94.60	82.30	65.32	95.50	81.90	89.70	92.40	72.40	57.50	77.80	80.90
Deformable R-FCN	87.30	81.40	63.60	90.40	81.60	74.10	90.30	75.30	71.40	75.50	79.10
Deformable Faster R-CNN	90.70	87.10	70.50	89.50	89.30	87.30	97.20	73.50	69.90	88.80	84.40
Multi-Scale CNN	99.30	92.00	83.20	97.20	90.80	92.60	98.10	85.10	71.90	85.90	89.60
RDAS512	99.60	85.50	89.00	95.00	89.60	94.80	95.30	82.60	77.20	86.50	89.50
R^2CNN++	100.00	89.41	97.22	97.00	83.15	87.54	99.17	99.40	74.51	90.10	91.75
Improved Faster R-CNN	99.58	88.13	93.78	97.88	85.76	93.06	97.67	97.44	68.97	93.52	92.31
Weighted Ensemble	100.00	97.80	69.82	96.80	93.10	96.81	99.70	100.00	90.30	93.40	94.20
YOLOv3	99.63	82.81	99.61	91.86	88.11	81.59	99.73	86.40	87.97	90.04	90.77
YOLOv3-Rot	99.92	89.40	99.62	92.71	90.25	85.25	99.15	89.73	92.24	92.74	93.10
YOLOv3-Att	100.00	91.60	99.93	94.70	91.97	86.62	97.52	94.80	93.18	94.59	94.49

TABLE III
EVALUATION RESULTS ON THE RSOD DATA SETS

Method	Aircraft	Oil tank	Overpass	Playground	mAP
R-P-Faster R-CNN	70.84	90.19	78.74	98.09	84.47
Deformable R-FCN	71.50	90.26	81.48	99.53	85.70
Deformable R-FCN and arcNMS	71.87	90.35	89.59	99.88	87.92
Faster R-CNN	71.90	90.90	100.00	100.00	90.70
SE block	80.50	90.90	90.90	100.00	90.60
DCFPN	87.64	97.12	93.10	100.00	94.47
RFN	80.10	99.40	100.00	100.00	94.90
Weight Ensamble	94.70	93.60	91.20	100.00	94.90
YOLOv3	97.14	96.98	93.15	99.93	96.80
YOLOv3-Rot	97.87	97.70	93.11	99.08	96.94
YOLOv3-Att	98.36	99.24	96.49	99.74	98.46

TABLE IV
EVALUATION RESULTS ON THE UCAS-ROD DATA SETS

Method	Plane	Car	mAP
YOLOv2	86.66	55.82	71.24
SSD	92.18	65.58	78.88
Faster R-CNN	95.00	83.00	89.00
R-FCN	97.60	89.30	93.50
Deformable R-FCN	97.60	91.70	94.70
A^2RMNet	99.24	94.65	96.94
YOLOv3	98.60	92.26	95.43
YOLOv3-Rot	99.27	93.91	96.59
YOLOv3-Att	99.32	95.87	97.55

the baseline model of YOLOv3 achieves the mAP of 90.77%. By using the GTM, YOlOv3-Rot improves the performance of YOLOv3 and gets the mAP of 93.10%. The improvement is mainly focused on the target category that contains less instance in the data set (i.e., basketball court and bridge) and target category that contain various rotation angles (i.e., ship). Moreover, the GCFM further improves the performance, especially for the classes of bridge and ship.

On the RSOD data set, we compared our methods with R-P-Faster R-CNN [42], Deformable R-FCN [66], Deformable R-FCN with arcNMS [66], Faster R-CNN [35], DCFPN [70], SEblock [71], RFN [12], and Weighted Ensemble [69]. Compared with these methods, our proposed model of YOLOv3-Att gets the highest mAP of 98.46%. This is a significant improvement over DCFPN (mAP 94.47%) and RFN (94.90%), which is mainly designed for the detection task where object orientation varies a lot. Notably, on the class aircraft, our method performs better than DCFPN and RFN at least 10%. We also make a comparison of YOLOv3, YOLOv3-Rot, and YOLOv3-Att. As shown in Table III, the baseline model of YOLOv3 achieves the mAP of 96.8%. The mAP of using GTM is higher, especially for the aircraft and oil tank. Furthermore, the GCFM helps the model achieve much better performance than that of only using GTM.

For the UCAS-ROD data set, we compared our methods with YOLOv2 [37], SSD [39], Faster R-CNN [35], R-FCN [65], Deformable R-FCN [51], and A^2RMNet [72]. The compared results are shown in Table IV. The detection accuracy of our proposed method is 97.55%, which goes beyond the other state of the arts. In particular, the mAP of the Car is significantly improved by using GCFM.

E. Computation Time Study

To show the effectiveness of the proposed method in practical cases, we also study the computational cost of the proposed

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TABLE V
INFERENCE SPEED COMPARISON OF DIFFERENT DETECTION METHODS

Methods	Mean Times (s)	FPS
RICNN	8.77	0.11
Faster R-CNN	0.16	6.25
Deformable R-FCN	0.20	4.98
Multi-Scale CNN	0.11	9.09
RDAS512	0.057	17.54
ORSIm Detector	0.21	4.72
FRIFB	0.88	1.14
YOLOv3-Att	0.082	12.13
YOLOv3	0.078	12.75

method. By adding the GFCM module, our proposed method YOLOv3-Att has about 2% more parameters compared with YOLOv3 that has about 61.6 million parameters. The main calculation in GFCM is matrix multiplication, which is fast and space-efficient in practice, since it can be implemented efficiently using GPU. To quantitatively evaluate inference speed, we run YOLOv3-Att and YOLOv3 on our machine with the input size 416×416 to compare the speed. As a reference, we also show the inference speed of some other competitors, and Table V shows the mean running time per image and the frames per second (FPS) of these approaches. From Table V, the detection speed of YOLOv3 is faster than most of the compared methods. YOLOv3-Att runs slightly slower than YOLOv3 (12.13 versus 12.75 FPS), which means that YOLOv3-Att can achieve a relatively high AP without large time consumption.

F. Ablation Study

To evaluate how the components of the proposed approach influence the performance, we conduct several ablation experiments on the NWPU VHR-10 data sets. A consistent setting is imposed on all the experiments, unless when some components or structures are examined. We study the impact of different configurations of the GTM and GCFM in four aspects. Especially, for the GTM, we study the type of transformation. For the GCFM, we study the dimension of the linear projection vector, the number of heads of attention, and the type of position embedding. The detection mAPs on validation set are shown in Table VI. The default setting of each aspect is marked with an asterisk.

1) Type of Transformation: The type of transformations used in GTM may have different influences on the performance. We try different combinations of transformations for the GTM: only rotation transformation (RT), only flip transformation (FT), and rotation and flip transformation (RT + FT). The results in Table VI show that RT seems to contribute more to the performance than FT. However, RT+FT achieves the best performance.

2) Dimension of Linear Projection Vector: In GCFM, the dimension of linear projection vector d_v is determined by the ratio r, as shown in Section II-C. Different values of r are conducted for comparison in this experiment. Moreover, in order to show how important the linear projection is to the performance, we also show the result of removing the linear projection operation, which is denoted as $r = 0$. From Table VI, we observe that the best performance is achieved when $r = 4$, and the performance gain is about 1%. It can be inferred that the linear projection has a positive effect on performance. It is worth noticing that the performance does not always rise with d_v increasing. This is because d_v equals the channels of the final fused feature map with global contextual information. Larger d_v means that the contextual information will take up more information in the feature. However, too much contextual information may have an adverse impact on location accuracy.

TABLE VI
RESULT OF ABLATION STUDY ON THE NWPU VHR-10 DATA SET

	Method	mAP
Type of transformation	None	91.03
	RT	93.08
	FT	92.11
	RT+FT*	94.58
Dimension of linear projection vector	r = 1	93.96
	r = 2	94.35
	r = 4*	94.58
	r = 8	94.20
	r = 16	93.77
Number of heads of attention	h = 1*	94.58
	h = 2	94.21
	h = 4	93.95
	h = 8	93.58
	h = 16	93.91
Type of position embedding	None	94.23
	Learned*	94.58
	Fixed	94.37

3) Number of Heads of Attention: We test the performance of using multihead attention in GCFM. Multihead attention is an advanced implement of attention mechanism. It is equivalent to perform an attention fusing layer with different learned linear projections h times in parallel, yielding h output values that are then concatenated to get the final output. The result in Table VI shows that $h = 1$ achieves the best performance.

4) Type of Position Embedding: As pointed in Section II-C, the position embedding has two choices: fixed and learned. We test the two types of position embedding and present the result of removing position embedding for comparison. From the result, learned embedding achieves the best performance. Therefore, learned embedding is recommended as the position embedding operation.

IV. ANALYZE OF THE RESULT AND MODEL

In this section, we analyze the effects of the proposed GCFM module on detection performance. For the purpose of validating the effectiveness of the spatial attention mechanism in the GCFM module, we demonstrate some detection results of YOLOv3-Rot and YOLOv3-Att on three data sets in Figs. 3–5. As shown in these figures, the GCFM could help model resulted in a better performance in detecting different kinds of objects. First, the detection results show that more objects are detected by using GCFM, which means that the global contextual information could help improve the confidence

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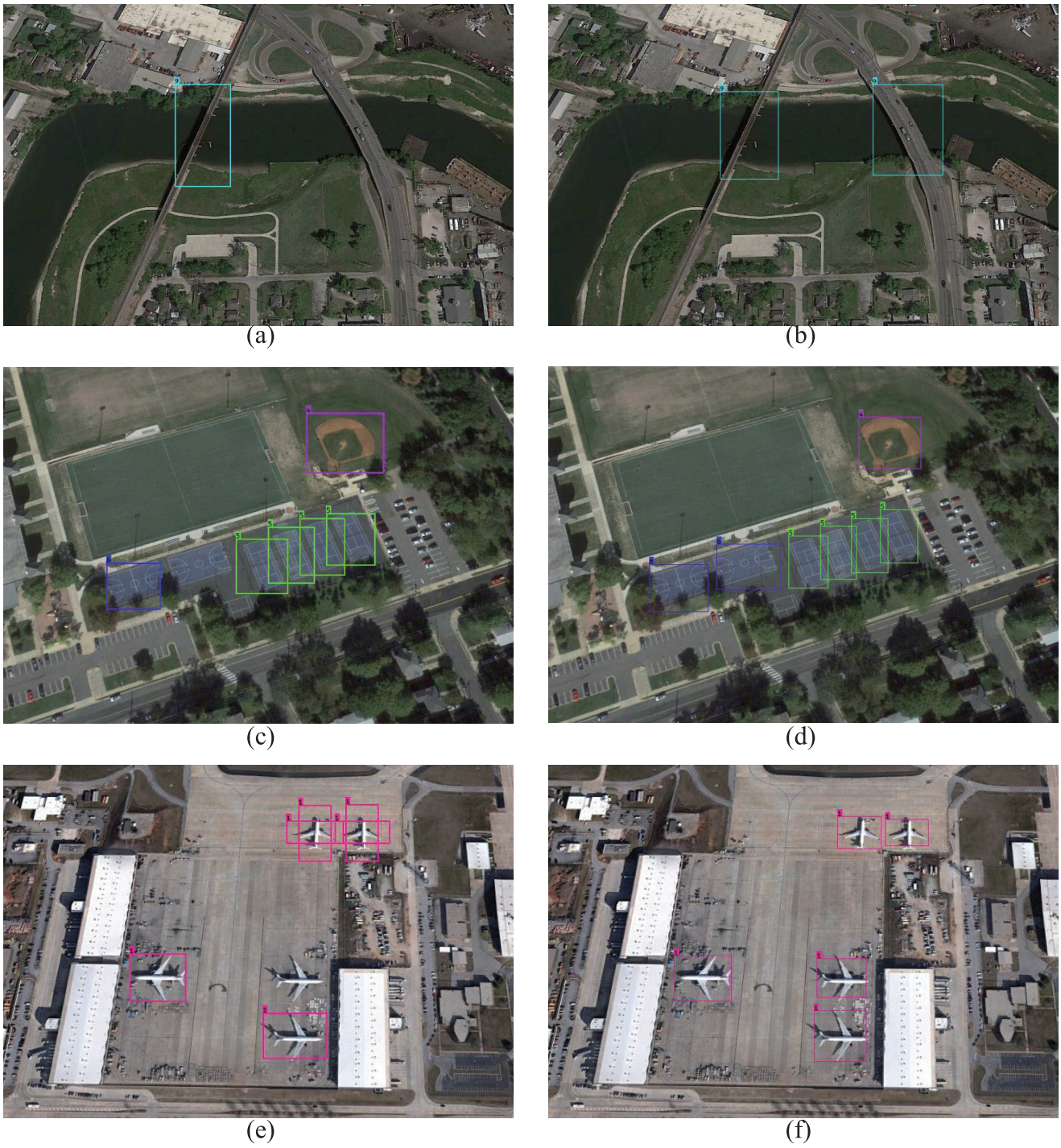


Fig. 3. Detection results of YOLOv3-Rot and YOLOv3-Att on the NWPU-VHR 10 data sets. (a), (c), and (e) Results of YOLOv3-Rot. (b), (d), and (f) Results of YOLOv3-Att.

score of the object. Second, the redundant bounding boxes could be eliminated by GCFM. As shown in Fig. 3(e), more than one bounding boxes are predicted by YOLOv3-Rot to locate the target of the two airplanes in the top right of the image. Nevertheless, if we applied GCFM, as shown in Fig. 3(f), the redundant bounding boxes are eliminated. Similar phenomena can be observed in Fig. 4(c) and (d). Furthermore, location accuracy is improved. The bounding boxes that predicted by using GCFM are more consistent with the actual object, as shown in Fig. 4(e) for detecting the playground and Fig. 3(f) for detecting the tennis court.

Such an obvious improvement of YOLOv3-Att should be due to the fact that the embedded attention mechanism can utilize the global contextual information in the image. By using GCFM, the receptive field for every anchor box is extended to the whole image. To better understand what GCFM do,

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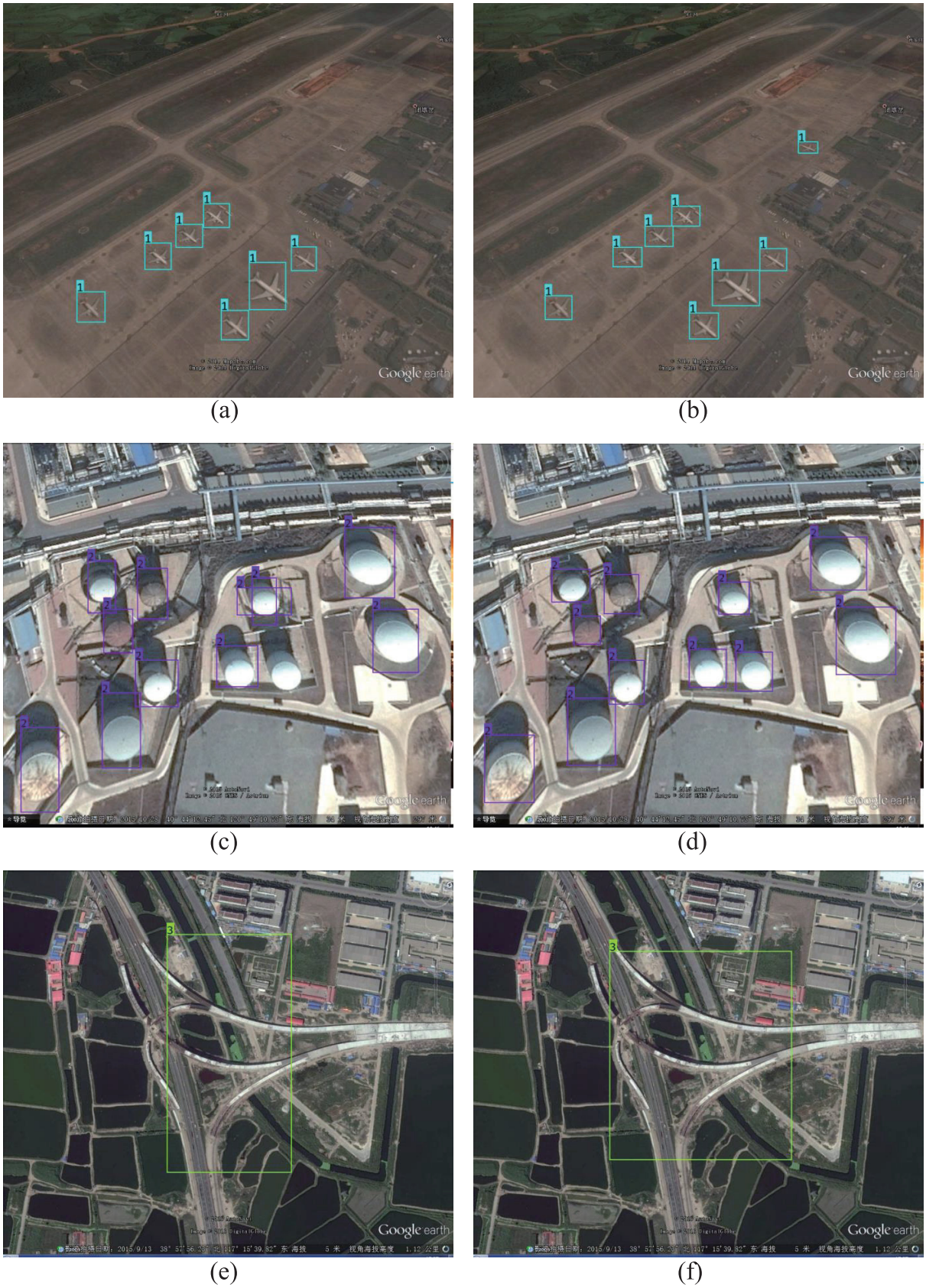


Fig. 4. Detection results of YOLOv3-Rot and YOLOv3-Att on the RSOD data sets. (a), (c), and (e) Results of YOLOv3-Rot. (b), (d), and (f) Results of YOLOv3-Att.

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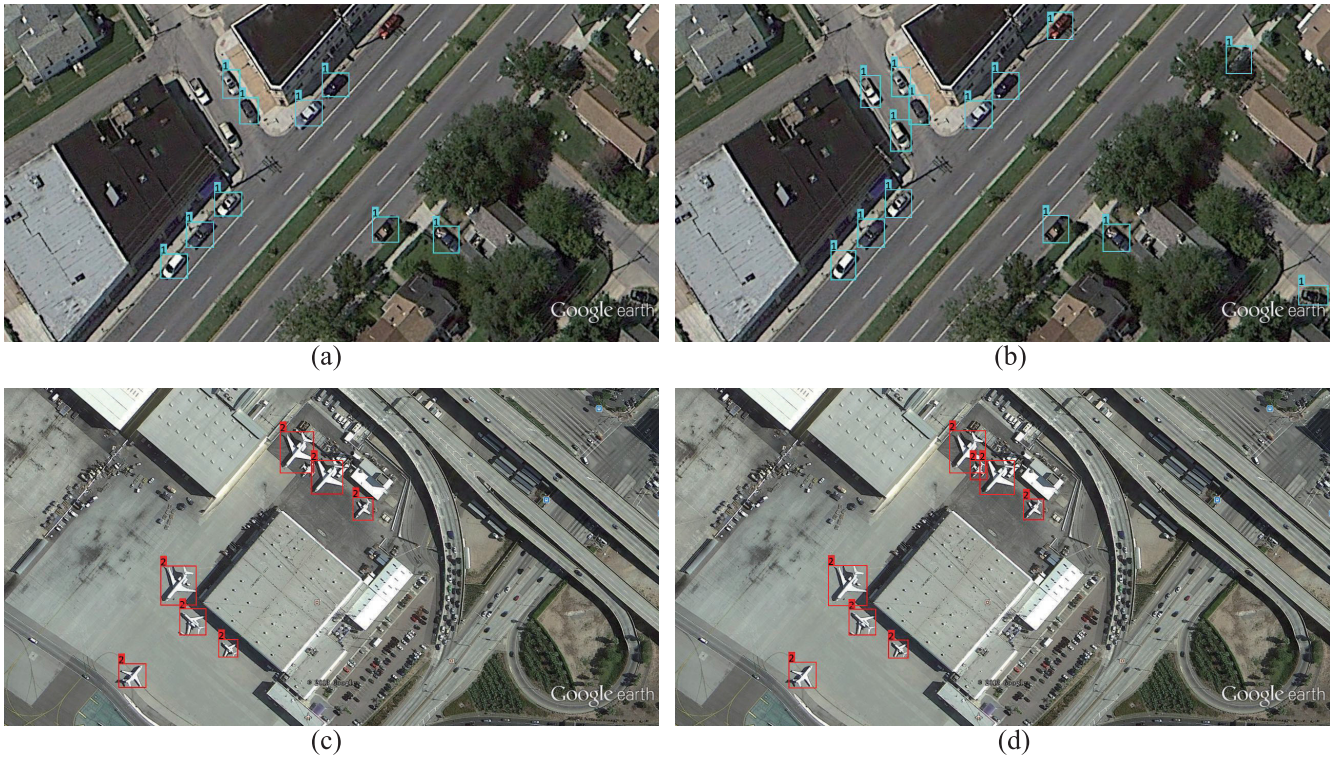


Fig. 5. Detection results of YOLOv3-Rot and YOLOv3-Att on the UCAS-AOD data sets. (a) and (c) Results of YOLOv3-Rot. (b) and (d) Results of YOLOv3-Att.

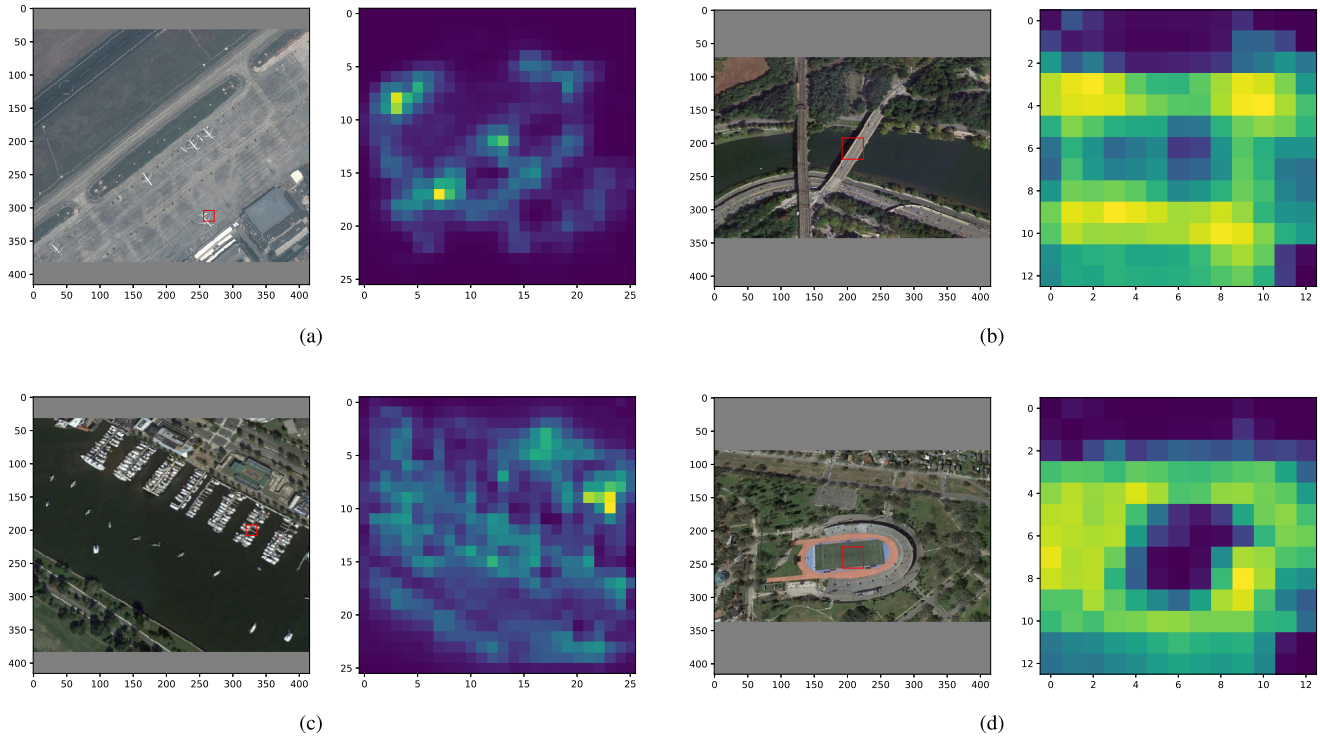


Fig. 6. Visualization of the attention weights learned by YOLOv3-Att. The attention weights of (a) airplane, (b) bridge, (c) harbor, and (d) playground.

Fig. 6 shows some attention weights learned by the YOLOv3-Att on the NWPU VHR-10 data set. The attention weights reflect the relative importance of positions at a feature map that the model utilized when detecting an object at the current position. As mentioned earlier, for each spatial position (i, j) in the feature map X^l_{pos}, the corresponding attention weight

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$A(i, j)$ is a $w_l \times h_l$ dimensional vector. Each element in $A(i, j)$ reflects the contribution of that element make to detect the object in (i, j). For better visualization, we resize the vector $A(i, j)$ to a 2-D $w_l \times h_l$ matrix and display the matrix by heatmap. In each subfigure of Fig. 6, the left figure is the corresponding input image. The right figure shows the attention weight heatmap of the position that is marked in the red rectangle of the input image. The color of the blocks in each heatmap represents the magnitude of attention weight.

In Fig. 6, we can see that, when the model detects objects, the contextual information of different areas is utilized. Fig. 6(a) shows the attention weight of an airplane. To detect the airplane, the model focuses on the area of the runway, the buildings, and the airfield, which are far away from the airplane. Fig. 6(b) shows the model uses riverbank as the contextual information when detecting a bridge. In Fig. 6(c), when detecting a harbor, the model focuses on the riverbank near the harbor and uses the river surface as the contextual information. When detecting a large object such as a playground in Fig. 6(d), the model focuses on the surrounding area of the playground.

The attention weight heatmaps in Fig. 6 show that the model focuses on different areas that are determined by the attention weight for detecting different objects. It is worth to mention that we use the concatenated feature as the input of the prediction module. Thus, in addition to those positions that have large attention weights, the model always pays attention to the position that the object is located. From the view of the receptive field, the receptive field that is used for detection in the traditional one-stage objection detection model is the local area around the current position. Our model extends the receptive field to the whole scale of the image and could improve the detection performance.

V. Conclusion

In this article, we propose a novel method that includes GTM and a global contextual fusion module for addressing the fast accuracy detection problem in remote sensing images. In particular, the GTM combines the rotation and flip transformation to address the orientation variation of the objects in remote sensing images. The global contextual fusion module uses the attention mechanism to learn the association of different locations in the feature map and fuse the features by attention weights to introduce contextual information both locally and globally. Considering the complex distribution of geospatial objects and the low efficiency of the current two-stage-based detection methods for remote sensing images, the GTM and global contextual fusion modules are considered and combined in the YOLOv3 object detection model.

The proposed framework presents a convincing performance on three publicly available remote sensing image object detection data sets. The quantitative evaluation results indicate that the proposed method excels state-of-the-art benchmarks for object detection in mAP. A detailed investigation confirmed that the added global contextual fusion module helps the model achieve better performance because it could utilize the global contextual information to improve the confidence score and generate more accurate results than the traditional method.

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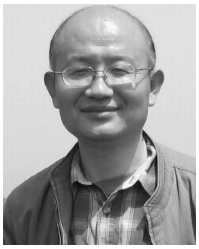
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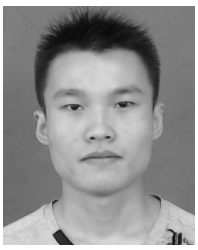
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